

MATHEMATICAL ANALYSIS / ACHIEVEMENT SETS / FRACTAL DIMENSION

EVERY INFINITE ACHIEVEMENT SET SPLITS INTO TWO

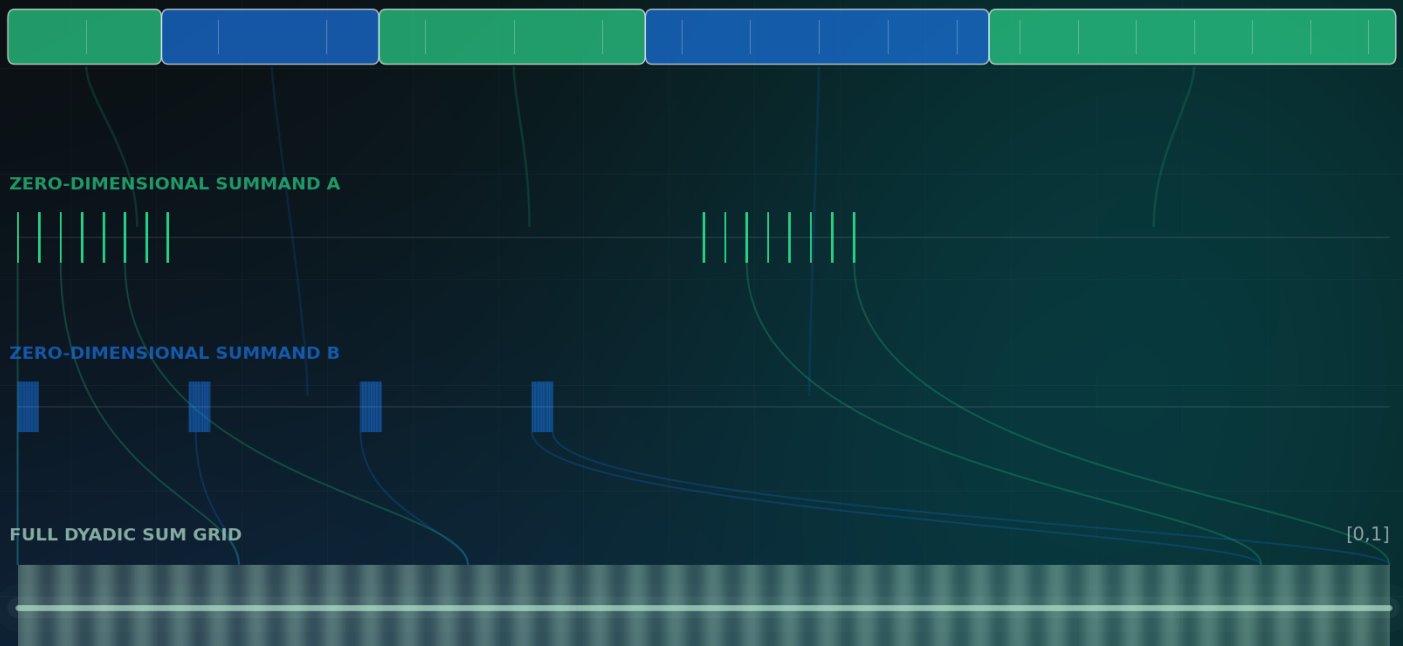
A zero-dimensional decomposition theorem and a candidate affirmative solution to Problem 16

PROPOSED DECOMPOSITION

$$\mathcal{E}(a) = \mathcal{E}(a|_A) + \mathcal{E}(a|_B),$$

$$\dim_{\text{H}} \mathcal{E}(a|_A) = \dim_{\text{H}} \mathcal{E}(a|_B) = 0$$

ALTERNATING DIGIT BLOCKS



Every Infinite Achievement Set Is a Sum of Two Achievement Sets of Hausdorff Dimension Zero

An alternating-block decomposition and a candidate affirmative solution to Problem 16

Metriq PRISM Laboratory

Program for Research in Intelligent Systems and Methods

A research laboratory of Metriq Foundation, Inc. • Florida, United States • metriq.org/prism

Research Preprint MF-MATH-2026-03 • Version 1.3 • 16 July 2026

Research status

This manuscript presents a candidate resolution of a published open problem. It has not undergone independent mathematical review or peer review. The proof is elementary and exact, but its novelty, the precise interpretation of the problem statement, and the absence of overlooked prior art require verification by specialists. The paper should be cited as a research preprint, not as an established theorem, until that review is complete.

Abstract

Let $a = (a_n)$ be an absolutely summable real sequence with infinitely many nonzero terms, and let

$$\mathcal{E}(a) = \left\{ \sum_{n=1}^{\infty} \varepsilon_n a_n : \varepsilon_n \in \{0, 1\} \right\}$$

be its achievement set. We construct a partition $\mathbb{N} = A \sqcup B$ into two infinite sets such that

$$\mathcal{E}(a) = \mathcal{E}(a|_A) + \mathcal{E}(a|_B) \quad \text{and} \quad \dim_{\text{H}} \mathcal{E}(a|_A) = \dim_{\text{H}} \mathcal{E}(a|_B) = 0.$$

The partition is formed from alternating finite blocks. After each block assigned to one subsequence, the next block is extended until the original series tail is super-exponentially small relative to the number of terms already assigned to the first subsequence. A standard cylinder cover then gives zero Hausdorff dimension at an infinite sequence of checkpoints. The same construction is applied symmetrically to the other subsequence. Both summands are in fact achievable Cantor sets. Subject to independent verification and the literature-status caveat above, this gives an affirmative answer to Problem 16 in the 2025 survey of Głab and Prus-Wiśniowski. We also give the explicit identity $[0, 1] = C_A + C_B$, where C_A and C_B are zero-dimensional achievement sets obtained by assigning factorial-length blocks of binary digits alternately.

CANDIDATE MAIN RESULT

For every infinite achievement set $\mathcal{E}(a)$ there are two achievement sets Y and Z of Hausdorff dimension zero such that

$$\mathcal{E}(a) = Y + Z.$$

The factors can be chosen as achievement sets of complementary subsequences of the original generating sequence.

MSC 2020. 40A05; 28A78; 54F45. **Keywords.** Achievement set; subsum set; Hausdorff dimension; algebraic sum; Cantor set; block decomposition.

1 Introduction

For an absolutely summable real sequence $a = (a_n)$, the *achievement set* (or *subsum set*) is the compact set of all subsums

$$\mathcal{E}(a) = \left\{ \sum_{n=1}^{\infty} \varepsilon_n a_n : \varepsilon_n \in \{0, 1\} \right\}. \quad (1)$$

A subset of the real line is called *achievable* if it is the achievement set of such a sequence. The subject links infinite series, additive structure, topology, and fractal geometry.

The 2025 survey by Głab and Prus-Wiśniowski records the following question [2]:

Problem 16 (Głab–Prus-Wiśniowski)

Can every infinite achievable set be decomposed into the sum of two achievable sets of Hausdorff dimension zero?

The question strengthens a theorem of Marchwicki, Nowakowski, and Prus-Wiśniowski: every infinite achievable set is the algebraic sum of two achievable Cantor sets of Lebesgue measure zero [3]. Zero Hausdorff dimension is substantially more restrictive than zero Lebesgue measure.

This paper proposes a direct affirmative solution. The key observation is that Hausdorff dimension can be forced to zero by making a subsequence tail super-exponentially small at infinitely many prefix lengths. Alternating blocks allow the intervening terms to be assigned to the other subsequence. Each factor therefore receives infinitely many terms, while each also benefits from arbitrarily long blocks assigned to its counterpart.

1.1 Outline of the argument

Enumerate the nonzero terms in inherited order as $b_1, b_2, \dots$ and write

$$\rho_N = \sum_{j>N} |b_j|$$

for their absolute tail. Partition the positions of this nonzero subsequence into consecutive blocks $I_1, I_2, \dots$, assigning odd blocks to A^* and even blocks to B^* . Suppose p_k positions have been assigned to A^* through block I_{2k-1} . Extend the following B^* -block until

$$\rho_{N_{2k}} \leq 2^{-p_k^2}.$$

The future absolute tail of the A^* -subsequence is then no larger than $2^{-p_k^2}$. Repeat with the roles exchanged. An achievement set generated by m initial terms and an absolute tail R_m is covered by at most 2^m tail cylinders. Hence, for every $s > 0$, the s -cost of the checkpoint cover is bounded by

$$2^{p_k} (2^{-p_k^2})^s = 2^{p_k - s p_k^2} \rightarrow 0.$$

This proves zero Hausdorff dimension for the first factor; the symmetric checkpoints prove the same for the second. Finally, the position partition is pulled back to the original indices, and zero terms are distributed arbitrarily.

1.2 Literature-status limitation

A targeted search completed on 16 July 2026 found the problem stated as open in the December 2025 survey and did not identify a later published or publicly indexed resolution. That search is not exhaustive and cannot establish publication priority. The construction should therefore be evaluated both for correctness and for possible equivalence to unpublished, non-indexed, or differently formulated prior work.

original sequence $a_1, a_2, \dots$ split into finite consecutive blocks

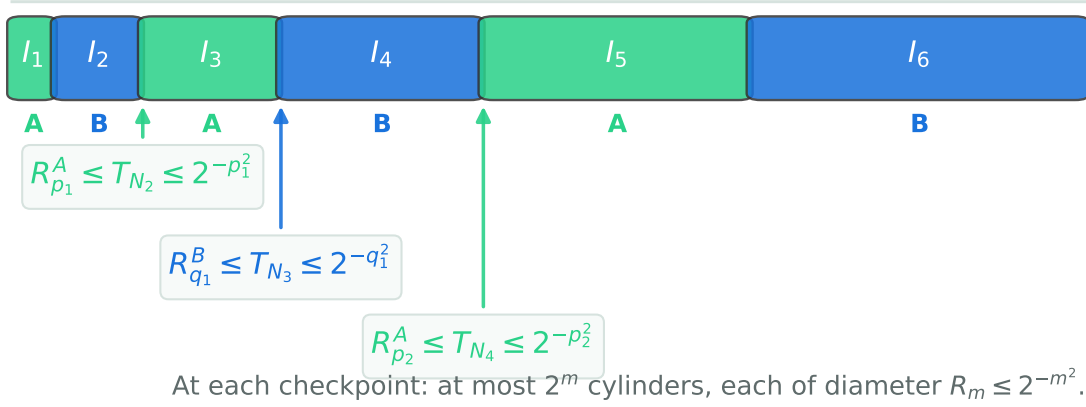


Figure 1: The alternating-block mechanism. A block assigned to one subsequence is lengthened until the original series tail is super-exponentially small relative to the number of terms already assigned to the other subsequence.

2 Preliminaries

2.1 Notation and Hausdorff dimension

For an absolutely summable real sequence $u = (u_n)$, write

$$R_m(u) = \sum_{n>m} |u_n|.$$

When the sequence is clear, we write R_m . For $F, G \subset \mathbb{R}$, their algebraic or Minkowski sum is

$$F + G = \{x + y : x \in F, y \in G\}.$$

For $s > 0$ and $\delta > 0$, define the s -dimensional Hausdorff content at scale δ by

$$\mathcal{H}_\delta^s(F) = \inf \left\{ \sum_i (\text{diam } U_i)^s : F \subseteq \bigcup_i U_i, \text{diam } U_i \leq \delta \right\}.$$

The Hausdorff measure is $\mathcal{H}^s(F) = \lim_{\delta \downarrow 0} \mathcal{H}_\delta^s(F)$, and

$$\dim_{\text{H}} F = \inf \{s > 0 : \mathcal{H}^s(F) = 0\}.$$

We use only this definition and the monotonicity of Hausdorff dimension; see Falconer for standard background [1].

2.2 A tail-cylinder covering lemma

Lemma 2.1 (Prefix-tail cover). *For every $m \geq 0$, the achievement set $\mathcal{E}(u)$ can be covered by at most 2^m sets of diameter at most $R_m(u)$. Consequently,*

$$\mathcal{H}_{R_m(u)}^s(\mathcal{E}(u)) \leq 2^m R_m(u)^s \quad (s > 0). \quad (2)$$

Proof. For each word $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m) \in \{0, 1\}^m$, let

$$s_\varepsilon = \sum_{j=1}^m \varepsilon_j u_j.$$

Every element of $\mathcal{E}(u)$ lies in one of the translated tail sets

$$s_\varepsilon + \mathcal{E}(u_{m+1}, u_{m+2}, \dots).$$

If x and y belong to the tail achievement set and arise from tail selections (ε_n) and (η_n) , then

$$|x - y| = \left| \sum_{n>m} (\varepsilon_n - \eta_n) u_n \right| \leq \sum_{n>m} |u_n| = R_m(u).$$

Thus the tail achievement set has diameter at most $R_m(u)$. There are at most 2^m translations. Summing the s th powers of their diameters gives (2). $\square$

Corollary 2.2 (Zero-dimensional checkpoint criterion). *Suppose there is a sequence $m_k \rightarrow \infty$ such that*

$$R_{m_k}(u) \leq 2^{-m_k^2}$$

for every k . Then $\dim_H \mathcal{E}(u) = 0$.

Proof. Fix $s > 0$. By [lemma 2.1](#),

$$\mathcal{H}_{R_{m_k}(u)}^s(\mathcal{E}(u)) \leq 2^{m_k - sm_k^2} \rightarrow 0.$$

Let $\delta > 0$ and $\eta > 0$. Because $m_k \rightarrow \infty$ and $R_{m_k}(u) \leq 2^{-m_k^2}$, choose k so large that

$$R_{m_k}(u) < \delta \quad \text{and} \quad 2^{m_k - sm_k^2} < \eta.$$

The cover supplied by [lemma 2.1](#) then has diameters below δ and total s -cost below η . Hence $\mathcal{H}_\delta^s(\mathcal{E}(u)) = 0$. Since $\delta > 0$ was arbitrary, $\mathcal{H}^s(\mathcal{E}(u)) = 0$. This holds for every $s > 0$, and therefore $\dim_H \mathcal{E}(u) = 0$. $\square$

Remark 2.3. The particular exponent m^2 is convenient rather than essential. It is enough to have checkpoint tails $R_{m_k}(u)$ satisfying

$$\frac{-\log R_{m_k}(u)}{m_k} \rightarrow \infty.$$

Any super-exponential decay relative to the subsequence prefix count gives the same conclusion.

3 The alternating-block partition

Let $a = (a_n)_{n \geq 1}$ be an absolutely summable real sequence with infinitely many nonzero terms. Write

$$J = \{n \in \mathbb{N} : a_n \neq 0\} = \{v_1 < v_2 < \dots\}$$

and let

$$b_j = a_{v_j} \quad (j \geq 1)$$

be the inherited subsequence of nonzero terms. We first partition the position set of b ; the resulting partition will be pulled back through $j \mapsto v_j$ in [theorem 4.1](#). Put

$$\rho_N = \sum_{j>N} |b_j|,$$

so $\rho_N \downarrow 0$.

We construct integers

$$0 = N_0 < N_1 < N_2 < N_3 < \dots$$

recursively. Choose any $N_1 \geq 1$. Once N_{2k-1} has been selected, define

$$p_k = \sum_{j=1}^k (N_{2j-1} - N_{2j-2}), \quad (3)$$

the total number of positions in the first k odd blocks. Choose $N_{2k} > N_{2k-1}$ so large that

$$\rho_{N_{2k}} \leq 2^{-p_k^2}. \quad (4)$$

This is possible because $\rho_N \rightarrow 0$.

After N_{2k} has been selected, define

$$q_k = \sum_{j=1}^k (N_{2j} - N_{2j-1}), \quad (5)$$

the total number of positions in the first k even blocks. Choose $N_{2k+1} > N_{2k}$ so large that

$$\rho_{N_{2k+1}} \leq 2^{-q_k^2}. \quad (6)$$

This completes the recursion.

Define the finite consecutive position blocks

$$I_j = \{N_{j-1} + 1, \dots, N_j\},$$

and set

$$A^* = \bigcup_{k=1}^{\infty} I_{2k-1}, \quad B^* = \bigcup_{k=1}^{\infty} I_{2k}. \quad (7)$$

Every block is nonempty, so A^* and B^* are both infinite. Let

$$y = b|_{A^*}, \quad z = b|_{B^*}$$

be the inherited subsequences.

Lemma 3.1 (Checkpoint tails). *For every $k \geq 1$,*

$$R_{p_k}(y) \leq \rho_{N_{2k}} \leq 2^{-p_k^2}, \quad (8)$$

$$R_{q_k}(z) \leq \rho_{N_{2k+1}} \leq 2^{-q_k^2}. \quad (9)$$

Proof. The first p_k terms of y are precisely the terms b_j whose positions lie in $I_1 \cup I_3 \cup \dots \cup I_{2k-1}$. The intervening block I_{2k} belongs to B^* . Consequently every term of y after its first p_k terms has original b -position greater than N_{2k} . Therefore

$$R_{p_k}(y) = \sum_{m > p_k} |y_m| \leq \sum_{j > N_{2k}} |b_j| = \rho_{N_{2k}} \leq 2^{-p_k^2}.$$

The proof for z is symmetric. Its first q_k terms occupy the even blocks through I_{2k} , while the following block I_{2k+1} belongs to A^* . Hence every later z -term has b -position greater than N_{2k+1} , and

$$R_{q_k}(z) \leq \rho_{N_{2k+1}} \leq 2^{-q_k^2}.$$

□

Proposition 3.2 (Both factors have dimension zero). *The achievement sets $\mathcal{E}(y)$ and $\mathcal{E}(z)$ satisfy*

$$\dim_{\text{H}} \mathcal{E}(y) = \dim_{\text{H}} \mathcal{E}(z) = 0.$$

Proof. Every odd block is nonempty, so $p_k \geq k$ and hence $p_k \rightarrow \infty$. Combining (8) with [corollary 2.2](#) gives $\dim_{\text{H}} \mathcal{E}(y) = 0$. Likewise every even block is nonempty, so $q_k \geq k$ and $q_k \rightarrow \infty$; (9) and [corollary 2.2](#) give $\dim_{\text{H}} \mathcal{E}(z) = 0$. $\square$

4 The decomposition theorem

Theorem 4.1 (Zero-dimensional decomposition). *Let $a = (a_n)$ be an absolutely summable real sequence with infinitely many nonzero terms. Then there is a partition $\mathbb{N} = A \sqcup B$ such that each of A and B contains infinitely many nonzero indices and, for the inherited subsequences $a|_A$ and $a|_B$,*

$$\mathcal{E}(a) = \mathcal{E}(a|_A) + \mathcal{E}(a|_B) \tag{10}$$

and

$$\dim_{\text{H}} \mathcal{E}(a|_A) = \dim_{\text{H}} \mathcal{E}(a|_B) = 0. \tag{11}$$

Proof. Use the notation of the preceding section. Pull the position partition $\mathbb{N} = A^* \sqcup B^*$ back to the original nonzero indices by setting

$$A_0 = \{v_j : j \in A^*\}, \quad B_0 = \{v_j : j \in B^*\}.$$

Let $Z = \{n : a_n = 0\}$. Choose any partition $Z = Z_A \sqcup Z_B$ and define

$$A = A_0 \cup Z_A, \quad B = B_0 \cup Z_B.$$

Then $\mathbb{N} = A \sqcup B$, and each part contains infinitely many nonzero indices. Because adjoining or deleting zero terms does not change an achievement set,

$$\mathcal{E}(a|_A) = \mathcal{E}(b|_{A^*}) = \mathcal{E}(y), \quad \mathcal{E}(a|_B) = \mathcal{E}(b|_{B^*}) = \mathcal{E}(z).$$

Thus (11) follows from [proposition 3.2](#).

It remains to prove the exact Minkowski-sum identity. For any selection $(\varepsilon_n)_{n \geq 1} \in \{0, 1\}^{\mathbb{N}}$, absolute convergence gives

$$\sum_{n=1}^{\infty} \varepsilon_n a_n = \sum_{n \in A} \varepsilon_n a_n + \sum_{n \in B} \varepsilon_n a_n.$$

The two terms on the right lie in $\mathcal{E}(a|_A)$ and $\mathcal{E}(a|_B)$, respectively. Hence

$$\mathcal{E}(a) \subseteq \mathcal{E}(a|_A) + \mathcal{E}(a|_B).$$

Conversely, choose arbitrary elements

$$x = \sum_{n \in A} \alpha_n a_n \in \mathcal{E}(a|_A), \quad y = \sum_{n \in B} \beta_n a_n \in \mathcal{E}(a|_B).$$

Define $\varepsilon_n = \alpha_n$ for $n \in A$ and $\varepsilon_n = \beta_n$ for $n \in B$. Absolute convergence permits the recombination

$$x + y = \sum_{n=1}^{\infty} \varepsilon_n a_n \in \mathcal{E}(a).$$

This proves the reverse inclusion and hence (10). $\square$

Corollary 4.2 (Candidate answer to Problem 16). *Subject to independent verification and the literature-status limitation in the introduction, [theorem 4.1](#) gives an affirmative answer to Problem 16 of Głqb and Prus-Wiśniowski [2].*

4.1 The summands are Cantor sets

Proposition 4.3. *The factors $\mathcal{E}(a|_A)$ and $\mathcal{E}(a|_B)$ in [theorem 4.1](#) are Cantor sets.*

Proof. Consider $\mathcal{E}(a|_A)$; the other factor is identical. It is compact as the continuous image of the compact product space $\{0, 1\}^{\mathbb{N}}$. It has no isolated points: fix a subsum represented by $(\varepsilon_n)_{n \in A}$ and a neighborhood of that sum. Because $a|_A$ contains infinitely many nonzero terms and $a_n \rightarrow 0$, choose a sufficiently late nonzero index $n \in A$ and toggle the single digit ε_n . The resulting sum differs by the nonzero quantity a_n and is arbitrarily close to the original sum. Finally, a subset of $\mathbb{R}$ with Hausdorff dimension zero contains no nondegenerate interval. Every connected subset of $\mathbb{R}$ is an interval, so all connected components are singletons. Thus the set is nonempty, compact, perfect, and totally disconnected; it is a Cantor set. $\square$

The proposed theorem therefore strengthens the earlier measure-zero decomposition theorem not only metrically but in a form preserving Cantor topology.

4.2 More than two factors

Corollary 4.4. *For every integer $m \geq 2$, every achievement set generated by an absolutely summable real sequence with infinitely many nonzero terms is the sum of m achievable Cantor sets of Hausdorff dimension zero, each generated by infinitely many nonzero terms.*

Proof. Apply [theorem 4.1](#) to obtain $\mathcal{E}(a) = \mathcal{E}(a|_A) + \mathcal{E}(a|_B)$ with both factors zero-dimensional and with infinitely many nonzero indices in each part. Enumerate the nonzero indices in B and distribute them cyclically among $m - 1$ disjoint infinite sets $B_1, \dots, B_{m-1}$; distribute any zero indices of B arbitrarily among those sets. Then

$$B = B_1 \sqcup \dots \sqcup B_{m-1},$$

and each B_j contains infinitely many nonzero indices. Splitting and recombining the coordinates exactly as in the proof of [theorem 4.1](#), with absolute convergence justifying the recombination, gives

$$\mathcal{E}(a|_B) = \mathcal{E}(a|_{B_1}) + \dots + \mathcal{E}(a|_{B_{m-1}}).$$

For every j , selecting zero on $B \setminus B_j$ shows that $\mathcal{E}(a|_{B_j}) \subseteq \mathcal{E}(a|_B)$. By monotonicity of Hausdorff dimension, each of these achievement sets has dimension zero. Each is a Cantor set by [proposition 4.3](#). Together with $\mathcal{E}(a|_A)$, this yields m zero-dimensional achievable Cantor summands. $\square$

5 An explicit binary model

The general proof is adaptive: the block lengths depend on the tail of the input series. For the classical binary series, a fixed factorial schedule gives a fully explicit example.

Let

$$a_n = 2^{-n}, \quad \mathcal{E}(a) = [0, 1].$$

Define

$$M_0 = 0, \quad M_j = \sum_{i=1}^j i!, \quad I_j = \{M_{j-1} + 1, \dots, M_j\}.$$

Let

$$A = \bigcup_{r \geq 1} I_{2r-1}, \quad B = \bigcup_{r \geq 1} I_{2r},$$

and put

$$C_A = \mathcal{E}((2^{-n})_{n \in A}), \quad C_B = \mathcal{E}((2^{-n})_{n \in B}). \quad (12)$$

Theorem 5.1 (Explicit interval decomposition). *The sets in (12) are zero-dimensional Cantor sets and*

$$\boxed{[0, 1] = C_A + C_B.}$$

Proof. Every $x \in [0, 1]$ has a binary subsum representation

$$x = \sum_{n=1}^{\infty} \varepsilon_n 2^{-n}, \quad \varepsilon_n \in \{0, 1\};$$

for $x = 1$, take the all-ones representation. Because every digit position belongs to exactly one of A or B , define

$$x_A = \sum_{n \in A} \varepsilon_n 2^{-n} \in C_A, \quad x_B = \sum_{n \in B} \varepsilon_n 2^{-n} \in C_B.$$

Then $x = x_A + x_B$, so $[0, 1] \subseteq C_A + C_B$. Conversely, any selections on the disjoint digit sets A and B combine to a single binary selection, whose sum lies in $[0, 1]$. Hence $C_A + C_B \subseteq [0, 1]$, and therefore $[0, 1] = C_A + C_B$.

It remains to prove the dimension statement. After the first r odd blocks, the number of assigned A -digits is

$$p_r = \sum_{j=1}^r (2j-1)!.$$

The next A -digit occurs after the entire even block I_{2r} , so

$$R_{p_r}((2^{-n})_{n \in A}) \leq \sum_{n > M_{2r}} 2^{-n} = 2^{-M_{2r}}.$$

For $r \geq 2$, an elementary induction gives

$$p_r \leq 2(2r-1)!, \quad M_{2r} \geq (2r)!,$$

and hence $M_{2r}/p_r \geq r \rightarrow \infty$. For every $s > 0$, the checkpoint cover from [lemma 2.1](#) has total s -cost at most

$$2^{p_r} (2^{-M_{2r}})^s = 2^{p_r - s M_{2r}} \rightarrow 0.$$

Its diameters also tend to zero, so the argument of [corollary 2.2](#) gives $\dim_H C_A = 0$.

Similarly, after the first r even blocks,

$$q_r = \sum_{j=1}^r (2j)! \leq 2(2r)!,$$

and every later B -digit occurs after $M_{2r+1} \geq (2r+1)!$. Thus

$$\frac{M_{2r+1}}{q_r} \geq \frac{2r+1}{2} \rightarrow \infty.$$

The corresponding checkpoint covers have total s -cost at most $2^{q_r - s M_{2r+1}} \rightarrow 0$, so $\dim_H C_B = 0$. Both generating subsequences contain infinitely many nonzero terms, and therefore both sets are Cantor sets by [proposition 4.3](#). $\square$

6 Robustness and extensions

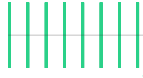
6.1 Absolute convergence is built into the theorem

[Theorem 4.1](#) is stated in the full absolutely convergent real-series setting. No positivity assumption is used. The prefix-tail cover controls the diameter of a signed tail by

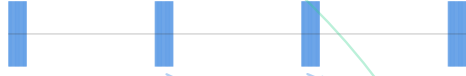
$$R_m(u) = \sum_{n > m} |u_n|;$$

the alternating block endpoints are chosen using the absolute nonzero-subsequence tail $\rho_N = \sum_{j > N} |b_j|$; and exact recombination follows from absolute convergence. Signed terms and zero terms are therefore part of the formal theorem rather than a separate extension.

finite prefix of C_A (digits from odd blocks)



finite prefix of C_B (digits from even blocks)



$C_A + C_B$: every 2^{-16} dyadic cell is attained



Disjoint binary digit positions imply exact addition without carries.

Figure 2: A finite-prefix view of the explicit binary construction. The odd and even factorial-length blocks occupy disjoint digit positions. Their sum recovers every dyadic cell and, in the limit, the full interval.

6.2 Why sparse single-term selection is insufficient

A naive partition into two sparse index sets does not necessarily preserve all terms: both factors must together contain the entire generating sequence. The alternating-block construction solves this bookkeeping constraint. A long block assigned to B simultaneously absorbs all intervening terms and creates a small tail checkpoint for A ; the next long block reverses the roles. No term is discarded, and exact algebraic recombination is automatic.

6.3 Quantitative freedom

The targets $2^{-p_k^2}$ and $2^{-q_k^2}$ can be replaced by arbitrary positive functions $\phi(p_k)$ and $\psi(q_k)$ for which

$$\frac{-\log \phi(m)}{m} \rightarrow \infty, \quad \frac{-\log \psi(m)}{m} \rightarrow \infty.$$

The construction can therefore impose additional spacing constraints or prescribed minimum block lengths without affecting the zero-dimensional conclusion. This flexibility may be useful for demanding topological, arithmetic, or coding properties of the summands.

7 Proof audit and review priorities

The central proof consists of four independently checkable steps:

1. **Cover count.** After m generator terms, at most 2^m translated tail sets cover the achievement set, and each has diameter no larger than the absolute tail.
2. **Recursive feasibility.** Since the absolute tail of the nonzero subsequence tends to zero, each alternating block endpoint can be pushed far enough to meet its super-exponential target.

3. **Index alignment.** The A^* -tail after p_k terms begins beyond position N_{2k} of the nonzero subsequence, while the B^* -tail after q_k terms begins beyond N_{2k+1} .

4. **Exact recombination.** Complementary index selections identify $\mathcal{E}(a)$ with the Minkowski sum of the two subsequence achievement sets.

The most important questions for independent reviewers are:

1. Does the phrase “decomposed into the sum” in Problem 16 impose any convention beyond an exact Minkowski-sum identity of achievable sets?
2. Is an equivalent alternating-block proof already present in the literature under a different dimension criterion or terminology?
3. Does the prevailing definition of an infinite achievable set admit conditionally convergent generators? The present theorem is explicitly limited to absolutely summable real sequences; zero terms are handled in the proof.
4. Does the explicit factorial binary example duplicate a known construction that should be cited directly?

All four mathematical steps above are proved symbolically. The supplementary exact-arithmetic script checks finite binary recombination and the factorial checkpoint ratios. Those computations are consistency tests, not a substitute for proof.

Institutional disclaimer and computational research declaration

This manuscript is an exploratory research preprint issued by the Metriq PRISM Laboratory, an interdisciplinary research laboratory of Metriq Foundation, Inc. PRISM—the Program for Research in Intelligent Systems and Methods—conducts research involving computational, mathematical, scientific, and machine-assisted methods.

The work was developed as part of PRISM’s experimental research program in computationally assisted mathematics. Metriq Foundation, acting through the PRISM Laboratory, directed the research process, evaluated the resulting argument, determined the form and scope of the manuscript, and accepts responsibility for its publication as a candidate result.

Generative artificial intelligence and other computational tools were used as research instruments to support construction search, symbolic and numerical analysis, literature review, proof development, verification, and manuscript preparation. Their use does not constitute independent validation of the results presented.

This manuscript has not yet undergone independent peer review and should not be regarded as an established resolution of the stated problem unless and until its arguments have been examined and confirmed by qualified mathematicians. Independent verification, criticism, replication, and identification of errors are expressly invited.

Data, code, and reproducibility

No empirical dataset was used. The source package accompanying this preprint contains the LuaLaTeX manuscript, an exact-arithmetic verification script, the recorded verification output, the figure-generation script, and generated figures. The cover artwork is a formula-derived visualization of alternating binary digit blocks and their reconstructed dyadic sum grid; it is not a stock photograph.

Conflicts of interest

Metriq Foundation, Inc. declares no financial conflict of interest concerning the mathematical result presented in this preprint.

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